



EQUITY BANK PESALINK ISO20022 UPGRADE

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On-Premise

Outbound

Release Notes

Project Details

Application Name	outBoundMS
Project Type	Software Development
Project Version	release-prod-1.0
Name of Re-leaser	Kennedy N. Gatimu
File Mode of Delivery	Jar, conf, yml, lib, so
Prepared By	Kennedy N. Gatimu

Release History

Date	Version	Author	Comments
12/09/2022	release-prod-1.0	Kennedy Gatimu	Initial Release

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DEVELOPER API GUIDE DOCUMENT

PROJECT: Equity Pesalink ISO20022 Upgrade

Date: 12.09.2022

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Delivery Details

This build is available in the form a jar - **out-bound-ms-0.0.1-SNAPSHOT.jar**

All files included in the build are;

- i. out-bound-ms-0.0.1-SNAPSHOT.jar
- ii. pl2-outbound
- iii. wrapper
- iv. application-prod.yml
- v. lib folder
- vi. pl2-outbound.conf
- vii. wrapper.jar
- viii. libwrapper.so

Production Property File Update

The production application-prod.yml will be updated as follows;

1. Eureka property

Replace the localhost on the defaultZone, with the production server ip address;

```
eureka:
  instance:
    prefer-ip-address: true
  client:
    service-url:
      defaultZone: http://admin:${jhipster.registry.password}@localhost:8761/eureka/
```

2. Using jasypt online tool, encrypt the database username and password as follows;

Go to URL: <https://www.devglan.com/online-tools/jasypt-online-encryption-decryption>

Jasypt Online Encryption	Jasypt Online Decryption
Enter plain text to Encrypt	Enter the Jasypt Encrypted Text
Enter plain text to hash	Jasypt Encrypted Password
Select Type of Encryption	Select Action Type
One Way Encryption(Without Secret Text) ▾	Decrypt Password ▾
Secret Key To Be Used While Encryption	Secret Key Used during Encryption(Optional)
Enter Secret Key	Enter Secret Key
Encrypt	Match/Decrypt
Encrypted String:	Result:
Result goes here	Result goes here

From the site above

- i. Enter plain text to Encrypt (username, password)
 - ii. Select Two Way Encryption as the Type of Encryption
 - iii. Enter Secret Key To Be Used While Encryption (Here create a secret key)
 - iv. Encrypt the password
3. On the application-prod.yml, update the encrypted database details as follows;
- i. Replace the database name from **outBoundMS** to the production DB name
 - ii. Update the DB username and passwords as below

username: ENC(jasypt-Encrypted-username)

password: ENC(jasypt-Encrypted-password)

```
spring:
  datasource:
    driver-class-name: com.microsoft.sqlserver.jdbc.SQLServerDriver
    url: jdbc:sqlserver://sql-omnichannel-preprod-001.privatelink.database.windows.net:1433;database=outBoundMS;encrypt=true
    username: ENC(jasypt-Encrypted-username)
    password: ENC(jasypt-Encrypted-password)
  hikari:
```

- iii. Update the jasypt property with the Secret Key used during Encryption above

```
jasypt:
  encryptor:
    password: h3xzqqc73Tigy84pEiHQPPQap
```

4. If the current server port 8083 is not available, update it with a port number available in production, otherwise leave it as 8083.

NOTE: If the port is changed, inform the development team for a code rebuild.

```
server:
  port: 8083
```

5. Update the ipsl certificate credentials - password and alias with production credentials provided by sysadmin team.

```
encrypt:
  key-store:
    alias: equitybanktransport
    location: /u01/gag/plv2/certs/Equity-IPSL/EQUITYBANK_transport.p12
    password: 1EVi6d2cy0
```

6. Update production server ip address here

```
#server ip address
ip:
  address: 10.11.6.111
```

Deploying Production Files

1. Copy Files To Production Paths

Copy the provided files to the below production paths respectively;

- a) `/u01/gag/plv2/outbound/bin/out-bound-ms-0.0.1-SNAPSHOT.jar`
- b) `/u01/gag/plv2/outbound/bin/wrapper`
- c) `/u01/gag/plv2/outbound/bin/pl2-outbound`
- d) `/u01/gag/plv2/outbound/bin/config/application-prod.yml`
- e) `/u01/gag/plv2/outbound/bin/lib/lib` folder
- f) `/u01/gag/plv2/outbound/conf/pl2-outbound.conf`
- g) `/u01/gag/plv2/outbound/lib/wrapper.jar`
- h) `/u01/gag/plv2/outbound/lib/libwrapper.so`

2. Create environment at /u01/gag - Optional

- a) `cd /u01/gag`
- b) `source j11_env.sh`

3. Update yml file at target location with the supplied prod yml

- a) `cd /u01/gag/plv2/outbound/bin/`
- b) `jar uf pesa-link-business-ms-0.0.1-SNAPSHOT.jar config/application-prod.yml`

4. Restart services at /u01/gag/plv2/outbound/bin/

- a) `sudo ./pl2-outbound restart`

Post - Deployment Actions

Post - deployment actions will include;

- i. Table index creation
- ii. Table insert scripts

Table Index Creation

Run below queries to create indexes in the production database for the given table names.

1. Table: **banks**

```
CREATE INDEX index_bankId ON [dbo].[banks] (bank_id);
```

2. Table: **client_limit**

```
CREATE INDEX index_sourceAccount ON [dbo].[client_limit] (source_account);
```

3. Table: **param_configs**

```
CREATE INDEX index_configId ON [dbo].[param_configs] (config_id);
```

4. Table: **tpg_transactions**

```
CREATE INDEX index_messageId ON [dbo].[tpg_transactions] (msg_id);
```

```
CREATE INDEX index_endToEndId ON [dbo].[tpg_transactions] (end_to_end_id);
```

5. Table: **transaction_details**

```
CREATE INDEX index_originalRequestId ON [dbo].[transaction_details] (original_request_id);
```

```
CREATE INDEX index_endToEndMsgId ON [dbo].[transaction_details] (end_to_end_message_id);
```


Table Insert Scripts

1. Table **param_configs**

Note: This is not the final insert list for table **param_config**. We shall update once we get production data.

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('xmlns="urn:iso:std:iso:20022:tech:xsd:acmt.023.001.02"', 'ACMT023 NameSpace',
'schema.location.actmt0023','ENABLEB','xmlns="urn:iso:std:iso:20022:tech:xsd:acmt.023.001.02"');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('xmlns="urn:iso:std:iso:20022:tech:xsd:pacs.028.001.04"', 'Pacs028 NameSpace',
'schema.location.pac028','ENABLEB','xmlns="urn:iso:std:iso:20022:tech:xsd:pacs.028.001.04"');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('/api/v1/transaction/process', 'Finacle Debit Credit Url',
'ips.payment.status.url','ENABLEB','/api/v1/transaction/process');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('https://api.ips.pesalink.co.ke/iso20022/sync/v1/verification-request', 'IPSL Verification Request
url',
'ips.verification.request.url','ENABLEB','https://api.ips.pesalink.co.ke/iso20022/sync/v1/verification-reque
st');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('https://api.ips.pesalink.co.ke/iso20022/async/v1/credit-transfer', 'IPSL Credit Transfer url',
'ips.credit.transfer.url','ENABLEB','https://api.ips.pesalink.co.ke/iso20022/async/v1/credit-transfer');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('https://api.ips.pesalink.co.ke/iso20022/sync/v1/payment-status-request', 'IPSL Payment Status
Url',
'ips.payment.status.url','ENABLEB','https://api.ips.pesalink.co.ke/iso20022/sync/v1/payment-status-requ
est');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('https://102.69.239.82:8443/kba/webservices/v2/LookupDbWS?wsdl', 'Endpoint To IPSL MGW
Phone Lookup',
'ipsl.mgw.phone.lookup.url','ENABLEB','https://102.69.239.82:8443/kba/webservices/v2/LookupDbWS?
wsdl');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('EQUITY BANK', 'Username To IPSL MGW Phone Lookup',
'kba.inquire.username','ENABLEB','EQUITY BANK');
```

```
insert into [dbo].[param_configs] (config_bin_value, config_desc, config_id, config_status, config_value)
values ('123456', 'Password To IPSL MGW Phone Lookup', 'kba.inquire.password','ENABLEB','123456');
```

2. Table **banks**

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0001', 'KCB', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0002', 'Stanchart', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0003', 'ABSA', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0005', 'Bank of India', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0006', 'Bank of Baroda', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0007', 'NCBA', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0010', 'Prime Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0011', 'Coop Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0012', 'NBK', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0014', 'M-Oriental', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0016', 'Citi Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0017', 'Habib Bank AG Zurich', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0018', 'Middle East Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0019', 'Bank of Africa', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0023', 'Consolidated', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0025', 'Credit Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0026', 'Access Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0031', 'Stanbic Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0035', 'ABC Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0043', 'Eco Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0049', 'SPIRE Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0050', 'Paramount', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0051', 'Kingdom Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0053', 'Gt Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0054', 'Victoria Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0055', 'Guardian Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0057', 'I&M Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0059', 'Development Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0060', 'SBM', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0061', 'Housing finance', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0063', 'DTB', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0065', 'Mayfair Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0066', 'Sidian Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0068', 'Equity Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0070', 'Family Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0072', 'Gulf African Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0074', 'First Community Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0075', 'DIB Bank', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0076', 'UBA', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0078', 'KWFT', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0079', 'Faulu Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0099', 'Post Bank', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('9999', 'IPS (Instant Payment Switch)', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0086', 'Vooma', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0087', 'Pesapal', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0088', 'Interswitch', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0089', 'Stima Sacco', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0091', 'Flutterwave', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0092', 'iPay', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0093', 'KCBbulk', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0095', 'MFS', 'OFFLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0097', 'Telkom Kenya', 'ONLINE');
```

```
insert into [dbo].[banks] (bank_id, bank_name, bank_status)
values ('0037', 'Cellulant', 'OFFLINE');
```